Elim Christian College

Level 1 Physics, Earth and Space Science 1.4

92047 Demonstrate understanding of energy in a physical system

Credits: Five

Achievement	Achievement with Merit	Achievement with Excellence
Demonstrate understanding of energy in a physical system.	Explain energy in a physical system.	Analyse energy in a physical system.

You may find the following formulae useful.

$E_k = \frac{1}{2}mv^2$	$E_g = mg\Delta h$	$g = 10\ N\ kg^{-1}$	$W = Fd$
$Q = mc\Delta t$	$Q = mL$	$Energy\ efficieny = \frac{useful\ energy\ output}{energy\ input} \times 100\ \%$	
$P = VI$	$\Delta E = Pt$	$Energy\ efficieny = \frac{useful\ power\ output}{power\ input} \times 100\ \%$	

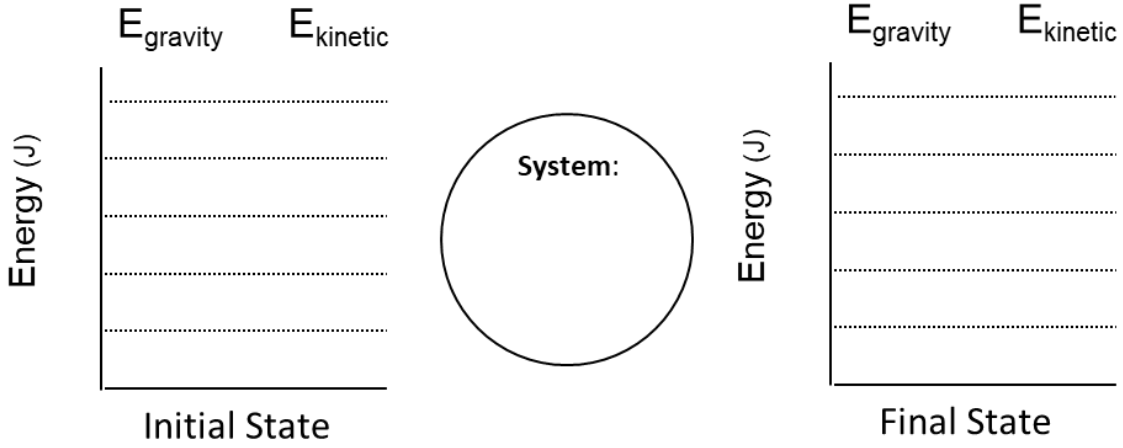
QUESTION ONE

(a) State the law of conservation of energy

Jamie plays with his football while he waits for the bus. He throws the ball vertically up. Consider the ball at two instants: the first instant (initial state) is when the ball is at its highest point, and the second instant (final state) is just before the ball hits the ground.

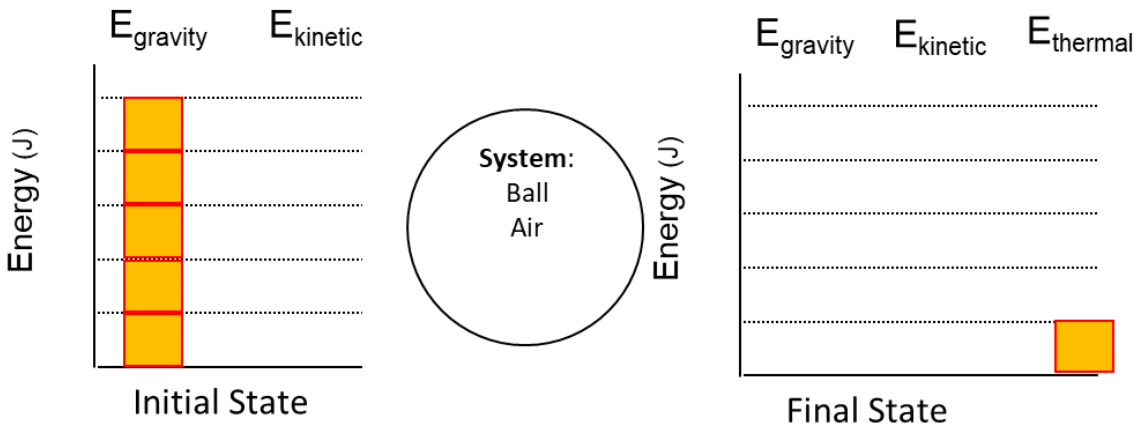


(b) Complete an energy bar chart for the ball at these two instants. (You may assume that the only forms of energy in this case are those shown on the energy bar chart.)



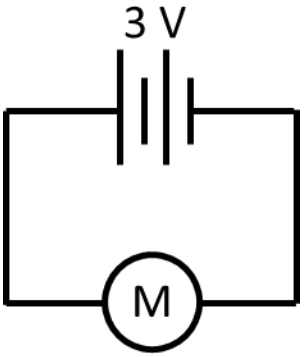
(c) The ball (with a mass of 0.4 kg) in the previous question reached a height of 6.25 m. Find the speed of the ball just before it hit the ground.

(d) A *different* second 0.4 kg ball is dropped from the same height (6.25 m). In this case the second ball coverts 5 J of its gravitational energy into thermal energy as it falls. Complete the energy bar chart below to show how the total energy is still conserved in this case, and use this energy bar chart to find the actual speed of the ball before it hits the ground.



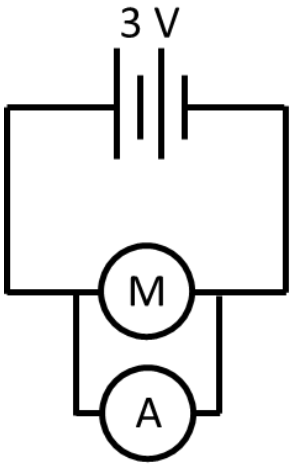
QUESTION TWO`

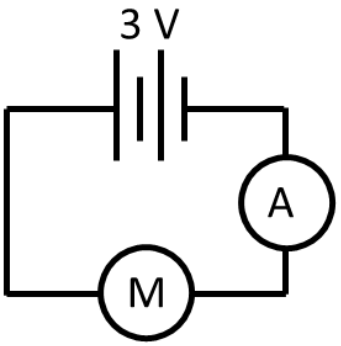
Pearl builds a small model electric elevator as a class project. The circuit has a battery pack and an electric motor as shown in the circuit diagram below.

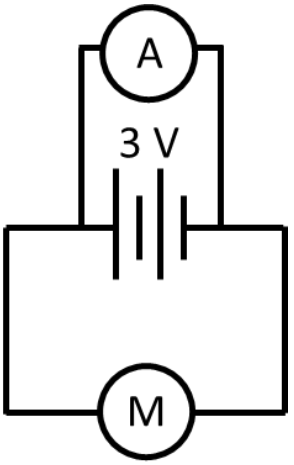


https://howtofunda.com/wp-content/uploads/2023/12/electric-lift-working-project.jpg

(a) Pearl wishes to add an ammeter to the circuit to measure the current. Circle the correct configuration in which the circuit diagram shows an ammeter wired up correctly.







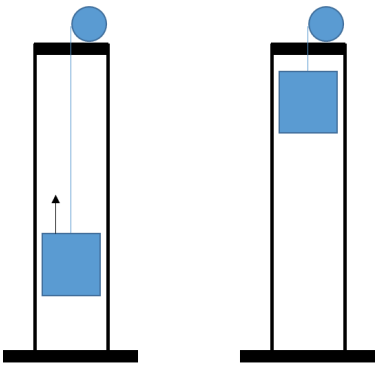
(Circle one)

(b) Pearl turns on the circuit and the elevator motor lifts a small mass for 10 s. If the current during this time is measured as 0.05 A, calculate the amount of electrical energy the battery outputs in this time.

(c) Pearl sets up the elevator again, this time using a larger battery, and the elevator lifts a 1 kg mass through a height of 0.1 m. The electrical energy being delivered by this battery is 1.25 J.

By first calculating the gravitational energy gained by the mass, calculate the energy efficiency of the electric elevator. (You may consider the initial state to be before the motor has started running, and the final state after the motor has finished running, so you may ignore kinetic energy in these two states).

(d) In terms of work and/or energy transfer, explain why each of the following statements is true. You could use an energy bar chart in your explanation, although this is not necessary.

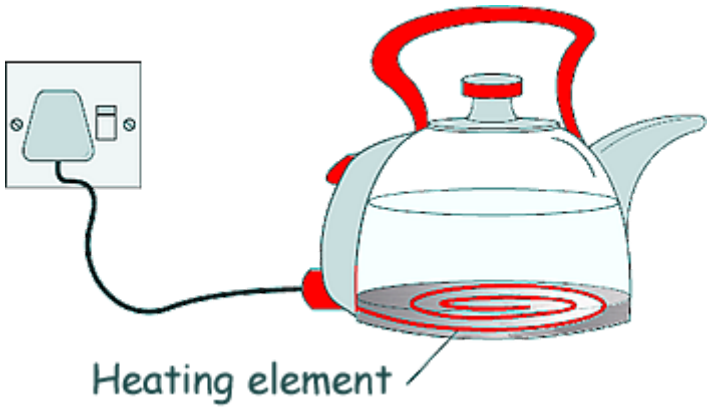


(i) Work is done on the mass as it is lifted to 0.1 m in height.

(ii) No work is done on the mass as it hangs stationary at the top of the lift.

QUESTION THREE

Jake has an electric kettle in his kitchen. He uses this kettle to heat up water. The kettle is well insulated and has a lid.



https://www.cyberphysics.co.uk/Q&A/KS4/heat/graphics/kettle.png

(a) Describe what happens to the water particles as the kettle is turned on and the water increases in temperature.

(b) The electric kettle has a heating element at the bottom, and yet all the water in the kettle eventually increases its temperature. Two heat transfer processes are mainly responsible for this.

(i) Name the heat transfer process that transfers thermal energy to the water touching the heating element.

(ii) Explain how the process of *convection* eventually transfers thermal energy to all the water in the kettle.

(c) The kettle delivers 1200 W of electric power in heating 0.8 kg water over a time of 3 minutes. If the specific heat capacity of water is 4200 J/kg/^{o}C, calculate the temperature change of the water.

(d) Jake repeats this process, but this time uses an uninsulated metal pot (with no lid), placed on a heating element, to heat the same mass of water as in question 3(c). The heating element below the pot delivers the same electric power to the pot and the water (as the heating element in the kettle), but after 3 minutes the temperature of the water has only increased by 50°C, which is less than the temperature increase using the kettle. Explain why using an uninsulated metal pot (with no lid) reduces the temperature rise (as compared to the kettle).